COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	**CPE212 / Automating Server Management**	**SECTION:**	**CPE31S2**
DATE PERFORMED:	**Oct 29 2026**	**DATE SUBMITTED:**	**Oct 29 2026**
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ACTIVITY NO. 12 **Finals Skills Exam**

1. PROCEDURE OUTPUT

Repository Link: https://github.com/qknlim/CPE212_Lim (under Final_Exam_Lim)
1. Create a repository and label it as "Final_Exam_Surname"
2. Clone your new repository in your VM
3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.
3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers
3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)
4.4 Change Motd as "Ansible Managed by <username>"
4. Push and commit your files in GitHub
5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)
5. For your final exam to be counted, please paste your repository link as an answer in this exam.
Note: Extra points if you will implement the said services via containerization.

To start: The enterprise service that I will be installing is Apache and Kibana as the monitoring tool on both Ubuntu and CentOS server.

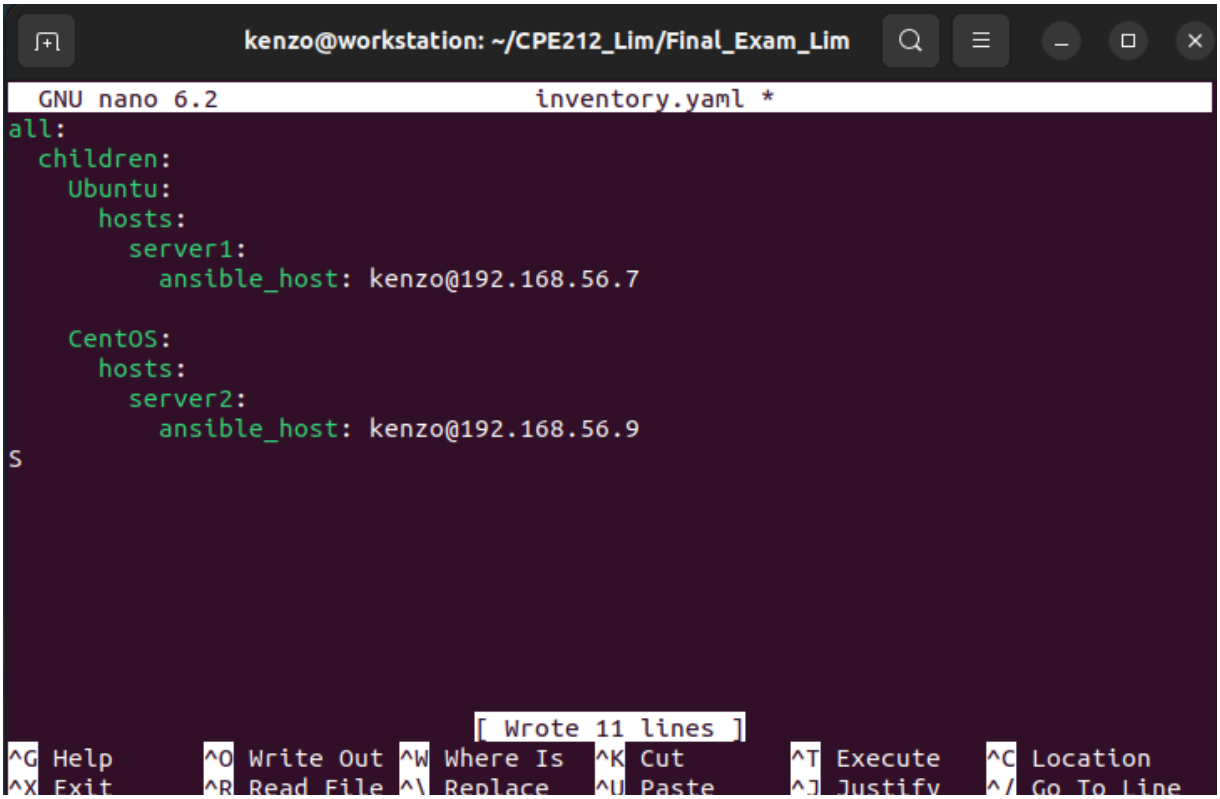


Figure 1.1 Inventory File

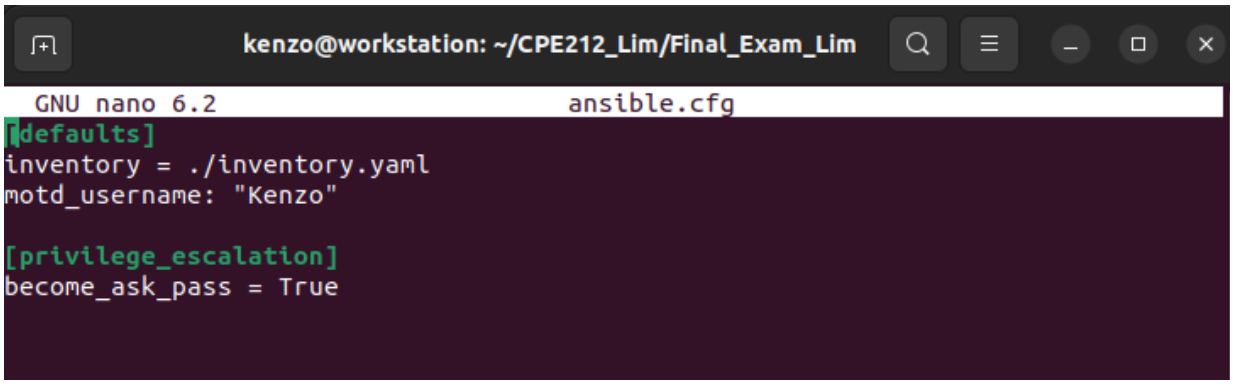


Figure 1.2 Config File

kenzo@workstation: ~/CPE212_Lim/Final_Exam_Lim

GNU nano 6.2 site.yaml

```
---

-  hosts: all
   become: true
   tasks:

    - name: install apache and php for Ubuntu servers
      tags: apache,apache2,ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: install apache and php for CentOS servers
      tags: apache,centos,httpd
      dnf:
        name:
          - httpd
          - php
        state: latest
      when: ansible_distribution == "CentOS"

    - name: start apache2 (Ubuntu)
      tags: apache, ubuntu, apache2
      service:
        name: apache2
        state: started
      when: ansible_distribution == "Ubuntu"

    - name: start httpd (CentOS)
      tags: apache, centos, httpd
      service:
        name: httpd
        state: started
      when: ansible_distribution == "CentOS"
```

Figure 1.3 Site.yaml

```yaml
- name: Apply common configurations across all ELK nodes
  hosts: all
  become: true
  roles:
    - common

- name: Install and configure Kibana
  hosts: all
  become: true
  roles:
    - kibana
```

Figure 1.4 Site.yaml continuation

kenzo@workstation: ~/CPE212_Lim/Final_Exam_Lim/roles/co...

GNU nano 6.2 main.yml

```
---
- name: Install prerequisites on Debian/Ubuntu
  apt:
    name:
      - apt-transport-https
      - gnupg2
      - openjdk-11-jre
    state: present
    update_cache: yes
  when: ansible_os_family == "Debian"

- name: Add Elastic GPG key on Debian/Ubuntu
  apt_key:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present
  when: ansible_os_family == "Debian"

- name: Add Elastic repository on Debian/Ubuntu
  apt_repository:
    repo: "deb https://artifacts.elastic.co/packages/8.x/apt stable main"
    state: present
    filename: elastic-8.x
  when: ansible_os_family == "Debian"

- name: Import Elastic GPG key on RHEL/CentOS
  rpm_key:
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    state: present
  when: ansible_os_family == "RedHat"

- name: Add Elastic repository on RHEL/CentOS
  yum_repository:
    name: elastic-8.x
    description: Elastic repository for 8.x packages
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: yes
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: yes
  when: ansible_os_family == "RedHat"
```

Figure 2.1 roles/common/tasks/main.yml

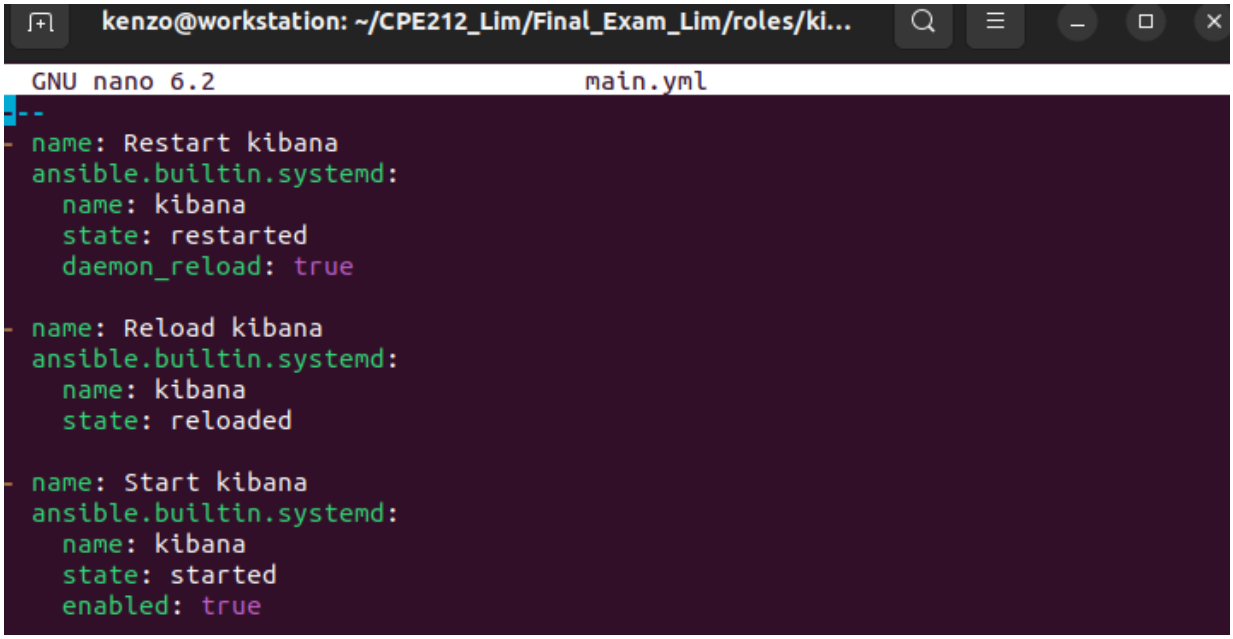


Figure 2.2 roles/kibana/handlers/tasks/main.yml

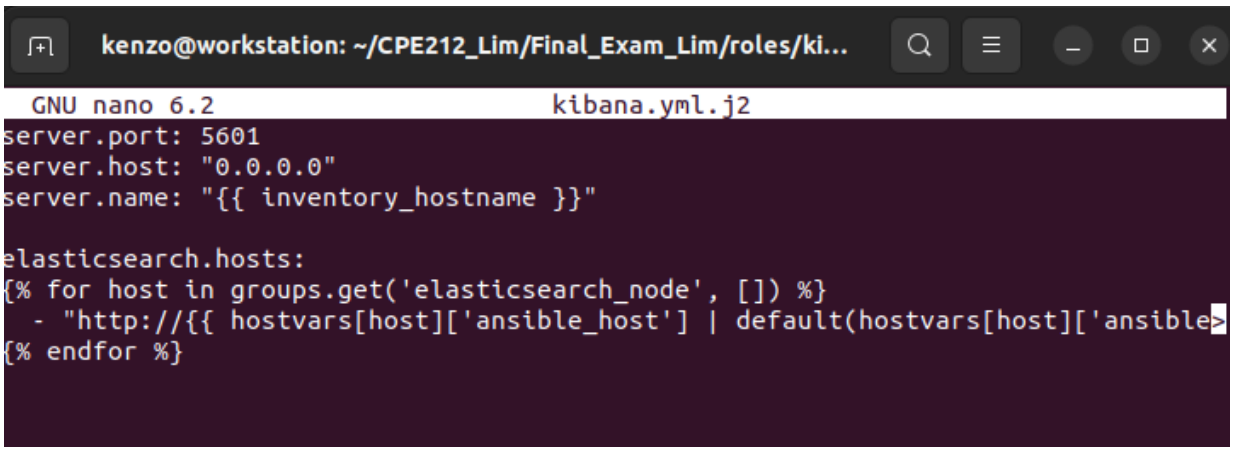


Figure 2.3 roles/kibana/templates/kibana.yml.j2

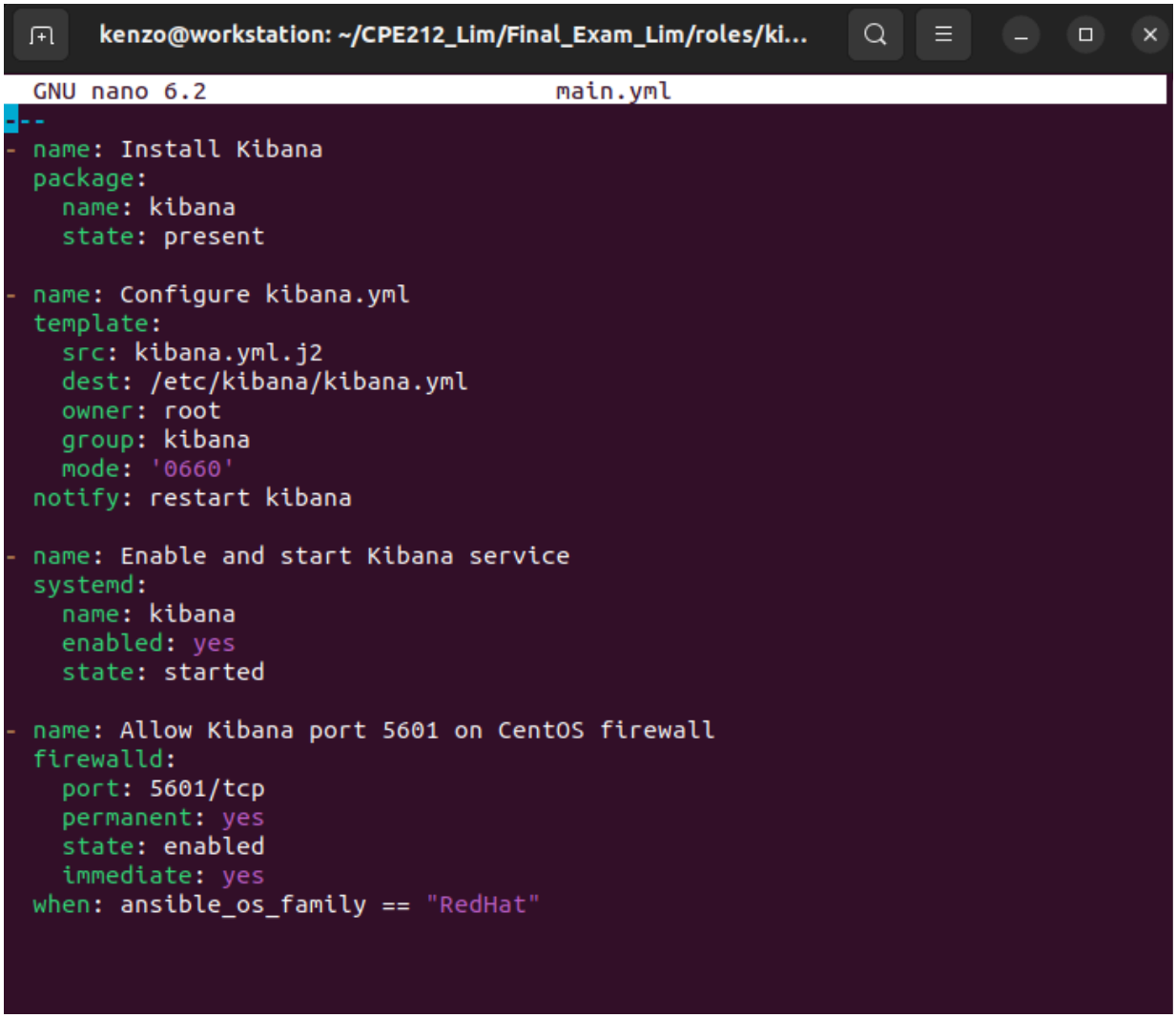


Figure 2.4 roles/kibana/tasks/main.yml

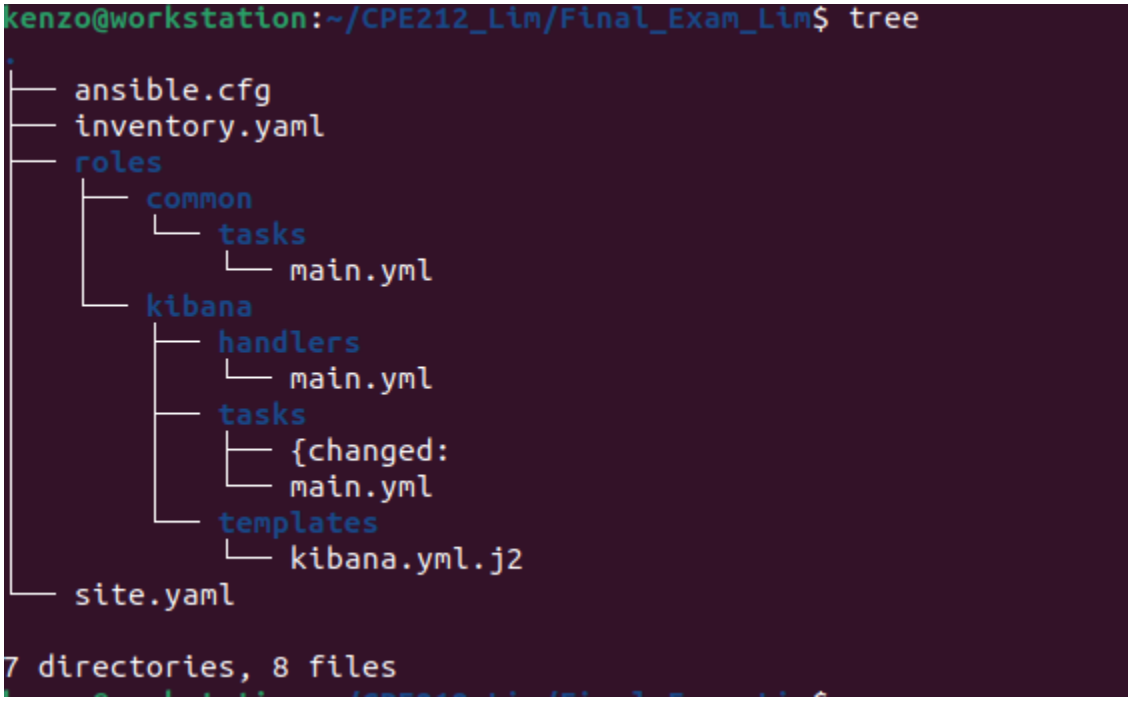


Figure 3.1 Tree overview of the directory

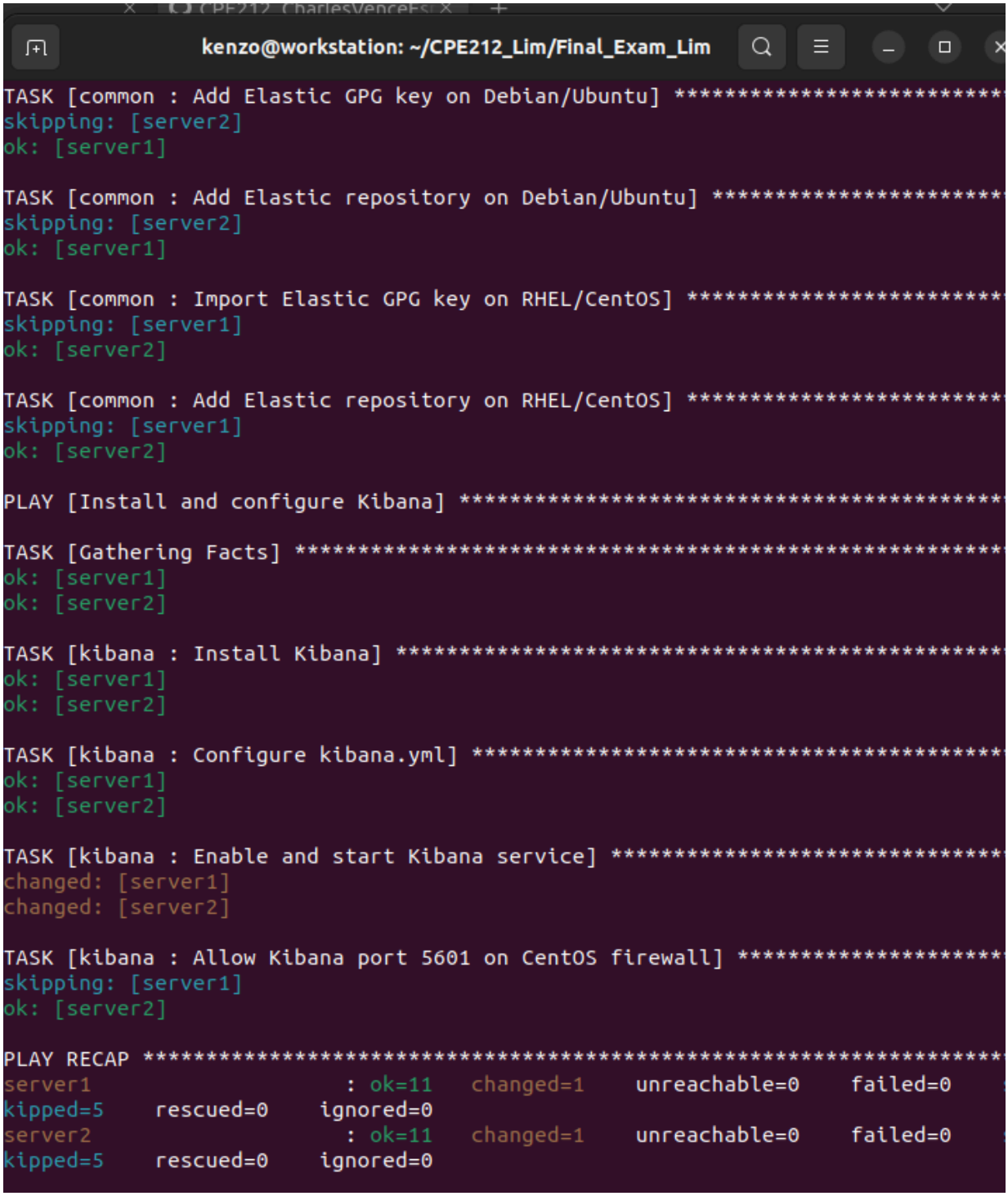


Figure 3.2 Result of the Playbook

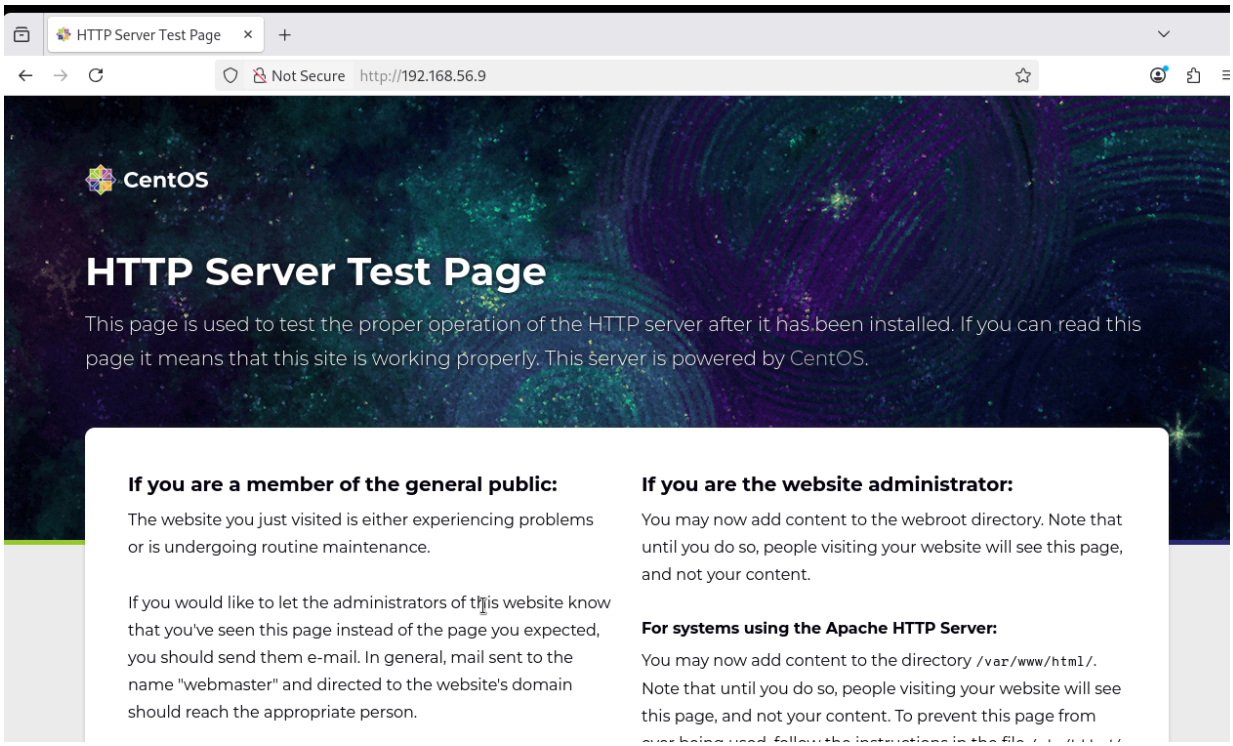


Figure 3.3 Verification for Apache2

```
kenzo@server1:~$ sudo systemctl status kibana
● kibana.service - Kibana
     Loaded: loaded (/lib/systemd/system/kibana.service; enabled; vendor preset: enabled)
     Active: activating (auto-restart) (Result: exit-code) since Thu 2026-10-29 10:25:12 PST; 2s ago
       Docs: https://www.elastic.co
    Process: 44435 ExecStart=/usr/share/kibana/bin/kibana (code=exited, status=78)
   Main PID: 44435 (code=exited, status=78)
        CPU: 6.220s
kenzo@server1:~$ curl -X GET "http://localhost:5601/api/status"
curl: (7) Failed to connect to localhost port 5601 after 0 ms: Connection refused
```

Figure 3.4 Verification for Kibana (FAIL)

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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